Class 10 Science – Chemistry Chapter 1

Chemical Reactions and Equations

Exam-oriented short notes • Definitions • Types of reactions • Important questions

1. Chemical Reaction

A chemical reaction is a process in which one or more substances change into new substances with different properties.

Common signs: change in colour, change in temperature, evolution of gas, formation of precipitate, or change in state.

2. Chemical Equation

A chemical equation represents a chemical reaction using chemical formulae and symbols. Example: 2H■ + O■ → 2H■O.

Balanced equation: The number of atoms of every element must be equal on both sides of the equation.

3. Major Types of Chemical Reactions

Combination reaction: Two or more substances combine to form a single product. Example: CaO + H■O → Ca(OH)■.

Decomposition reaction: One compound breaks into two or more simpler substances. Example: CaCO■ → CaO + CO■.

Displacement reaction: A more reactive element displaces a less reactive element. Example: Zn + CuSO■ → ZnSO■ + Cu.

Double displacement reaction: Two compounds exchange ions. Example: Na■SO■ + BaCl■ → BaSO■↓ + 2NaCl.

Redox reaction: Oxidation and reduction occur together.

4. Oxidation and Reduction

Oxidation: addition of oxygen, removal of hydrogen, or loss of electrons. **Reduction:** removal of oxygen, addition of hydrogen, or gain of electrons.

Example: CuO + H■ → Cu + H■O. Here CuO is reduced to Cu and H■ is oxidised to H■O.

5. Effects of Oxidation in Everyday Life

Corrosion: Slow damage of metals due to reaction with air/moisture, e.g. rusting of iron.

Rancidity: Oxidation of fats and oils causing unpleasant smell and taste. It can be slowed by refrigeration, airtight packing, antioxidants, or nitrogen flushing.

6. Important Exam Questions

Q1	What is a chemical reaction?
Q2	What is a balanced chemical equation?
Q3	Define combination and decomposition reactions with examples.
Q4	What is a displacement reaction? Give one example.
Q5	Differentiate between oxidation and reduction.
Q6	What is corrosion? How can it be prevented?
Q7	What is rancidity? Mention two methods to prevent it.

Quick Revision Tip: Learn the definitions, reaction types, one example of each, and the oxidation/reduction concepts first. These are frequent exam targets.